



Evaluation and benchmarking: Continuous testing for hallucinations, toxicity, bias, and factual accuracy.

Observability and monitoring: Track latency, cost, token usage, drift, and user feedback loops.

Safety, guardrails and policy enforcement: Apply content filters, grounding, red-teaming, and enterprise-grade safety layers.

Cost optimisation: Manage model routing, caching, batching, and hybrid model strategies.

Some of the open source LLMops tools/frameworks are: LangChain/LangGraph, LlamaIndex, Haystack, Agenta, Langfuse, Chroma, Milvus/Weaviate, and Arize Phoenix.

MLops has progressed from 'model operations' to 'intelligence operations', supporting AI systems that learn continuously, act autonomously, and operate safely at enterprise scale. CIOs need to treat MLOps as a core enterprise capability, not a data science accessory. MLOps tools and frameworks need to become the norm, especially for enterprises with sensitive data. They must invest in cross-skilling data engineers, ML engineers, and DevOps teams.

As generative AI models become mainstream, the future seems to be the emergence of LLMops. However, managing LLMs introduces new challenges like prompt management, fine-tuning pipelines, vector stores, and safety layers. MLOps teams must evolve into LLMops teams, integrating agentic workflows and safety governance. **END** 🐧

Acknowledgements

The authors would like to thank Tricon Solutions LLC and Gspann Technologies, Inc., for giving the required time and support in many ways when this article was being written.

By: Dr Gopala Krishna Behara and Narumanchi Sree Keerthi

Dr Gopala Krishna Behara is an enterprise architect at Tricon IT Solutions. He has around 28 years of experience in the IT industry.

Narumanchi Sree Keerthi is a senior software engineer at Gspann Technologies, Inc. She has around 5 years of experience in the IT industry.

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